

Geometry
Unit 9
Lesson 4 Homework

Name _____ **Answer Key** _____
Date _____
Period _____

Given the graph.

1. Draw the transformation of the triangle reflected across the y -axis.

See the solid red lines on the graph

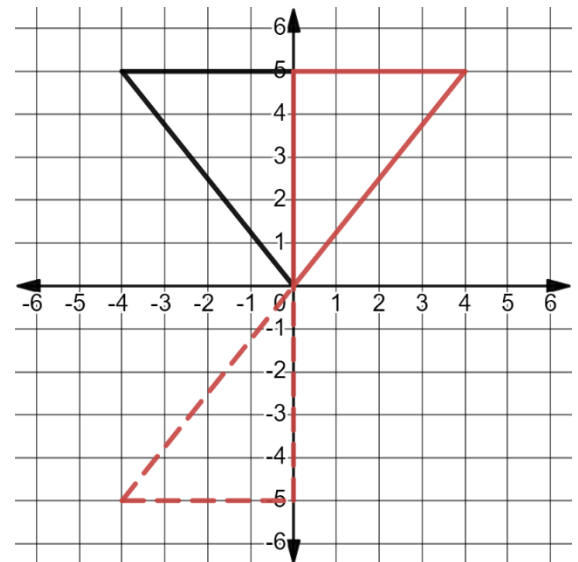
2. What three-dimensional shape would be created by rotating the triangle around the y -axis?

Cone

3. What are the base and height measurements of the three-dimensional shape, in units?

Base radius = 4 units

Height = 5 units



4. Draw the transformation of the triangle reflected across the x -axis in a different color.

See the dashed lines on the graph

5. What three-dimensional shape would be created by rotating the triangle around the x -axis?

Cylinder with a cone cutout

Given the circle.

6. Draw the transformation of the circle reflected across the y -axis.

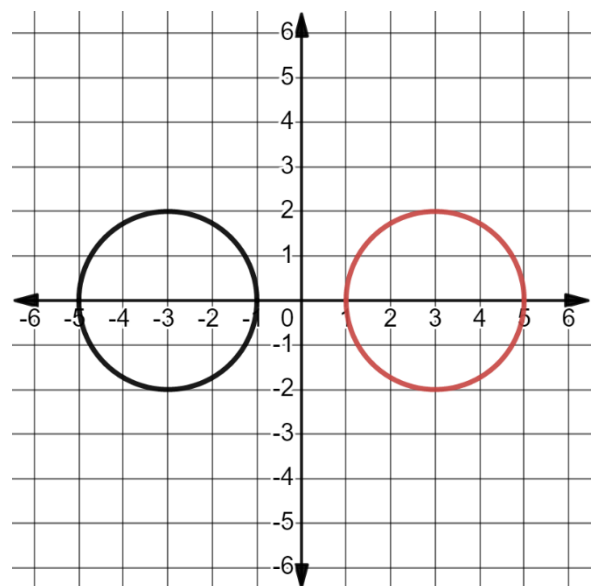
See red circle on graph

7. What three-dimensional shape would be created by rotating the circle around the y -axis?

Torus or donut

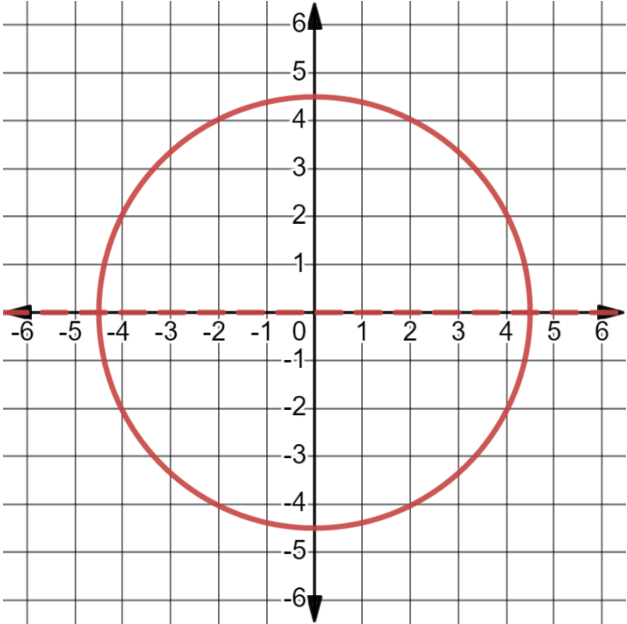
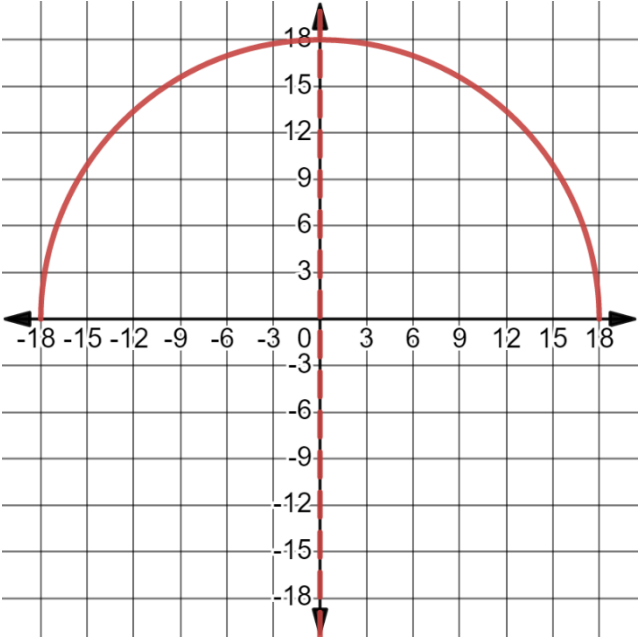
8. Name one real world object that exhibits the three-dimensional shape created.

Answers will vary. Examples: donut, washer



For questions 9-10, draw a two-dimensional figure and an axis of rotation that would result in the real-world object when the two-dimensional figure is rotated around the axis of rotation.

Answers will vary.

Image	Two-Dimensional Figure
9. A soccer ball with a radius of 4.5 inches 	 Answers will vary. Rotated around the x-axis
10. An umbrella with a radius of 18 inches 	 Answers will vary. Rotated around the y-axis